Truncation error of the signature representation of the Kalman–Bucy filter

Tamearon Mullings

Abstract

For a scalar Ornstein–Uhlenbeck signal observed in continuous time, the Kalman–Bucy filter mean is a linear functional of the signature of the time-augmented observation path (Cohen et al., 2023). Bounds on the error made by truncating this representation are left open there. In the constant-gain case we prove two-sided bounds for two choices of coefficients. With the exact series coefficients the error decays like $(\lambda T)^M/M!$ in the truncation level M. With coefficients chosen optimally at each level it is governed by the best L^2 polynomial approximation of the filter kernel, and decays like $(\lambda T/4)^M/M!$ up to a factor polynomial in M. Both results agree with simulations, and a Beneš example shows that the picture changes for nonlinear filters.

1 Introduction

The signature of a path, the collection of its iterated integrals, gives a linear basis for functionals of the path (Lyons, 1998). Cohen et al. (2023) show that the Kalman–Bucy filter (Kalman and Bucy, 1961) is a linear function of the initial estimate and of the signature of the time-augmented observation path, so that linear regression on signature features can recover the optimal linear filter. In practice the signature is truncated at a finite level M, and Cohen et al. (2023, App. H.3) note that bounding the resulting approximation error is an open problem. This note gives explicit, two-sided bounds in the simplest setting: a scalar Ornstein–Uhlenbeck (OU) signal with the filter started at its steady state.

2 Model and notation

Fix $T > 0$ and let W, V be independent standard Brownian motions. The signal and observation are

$$dX_t = -aX_t\,dt + \sigma\,dW_t, \qquad X_0 \sim \mathcal{N}(0, P_\infty), \tag{1}$$

$$dY_t = hX_t\,dt + \gamma\,dV_t, \qquad Y_0 = 0, \tag{2}$$

with $a, \sigma, h, \gamma > 0$ and X_0 independent of (W, V). Here P_∞ is the positive root of the algebraic Riccati equation

$$\sigma^2 - 2aP - \tfrac{h^2}{\gamma^2}P^2 = 0, \qquad \text{so} \qquad P_\infty = \frac{\gamma^2}{h^2}\Big(-a + \sqrt{a^2 + h^2\sigma^2/\gamma^2}\Big). \tag{3}$$

We write

$$K = \frac{P_\infty h}{\gamma^2}, \qquad \lambda = a + Kh = \sqrt{a^2 + h^2\sigma^2/\gamma^2}, \qquad k(u) = Ke^{-\lambda u}, \qquad c_X = \max\Big\{P_\infty,\ \frac{\sigma^2}{2a}\Big\}.$$

The second expression for λ follows from (3). For $f \in L^2[0,T]$ let $\|f\|^2 = \int_0^T f(u)^2\,du$, let $I(f) = \int_0^T f(T-s)\,dY_s$, let $\mathcal{P}_n$ be the polynomials of degree at most n, and let $E_n(f) = \min_{p\in\mathcal{P}_n}\|f - p\|$.

Signatures. The time-augmented path is $\hat{Y}_t = (t, Y_t)$, with letters $\mathtt{t}$ and $\mathtt{y}$. The signature coordinate of a word $w = w_1 \cdots w_n$ on $[0, s]$ is defined recursively by $S^{\varnothing}_{0,s} = 1$ and $S^{w_1 \cdots w_n}_{0,s} = \int_0^s S^{w_1 \cdots w_{n-1}}_{0,u}\, d\hat{Y}^{w_n}_u$ (Lyons, 1998). Only the words $\mathtt{y}\mathtt{t}^j$ (one $\mathtt{y}$ followed by j letters $\mathtt{t}$) are needed. For these, the only integral against Y has integrand 1, so Itô and Stratonovich conventions coincide.

3 The signature representation

Lemma 1. *For $j \geq 0$ and $s \in [0, T]$, $S^{\mathtt{y}\mathtt{t}^j}_{0,s} = \displaystyle\int_0^s \frac{(s-u)^j}{j!}\, dY_u$. In particular $S^{\mathtt{y}\mathtt{t}^j}_{0,T} = I\big(u^j/j!\big)$.*

Proof. For $j = 0$ both sides equal Y_s. For $j \geq 1$ the recursion gives $S^{\mathtt{y}\mathtt{t}^j}_{0,s} = \int_0^s S^{\mathtt{y}\mathtt{t}^{j-1}}_{0,u}\, du$, a pathwise Riemann integral. Iterating it with Cauchy's formula for repeated integration gives $S^{\mathtt{y}\mathtt{t}^j}_{0,s} = \int_0^s \frac{(s-u)^{j-1}}{(j-1)!}\, Y_u\, du$. Put $\phi(u) = -(s-u)^j/j!$, so $\phi'(u) = (s-u)^{j-1}/(j-1)!$. Since ϕ is deterministic and C^1, integration by parts (Protter, 2005, Ch. II) gives

$$\int_0^s \phi'(u) Y_u\, du = \phi(s) Y_s - \phi(0) Y_0 - \int_0^s \phi(u)\, dY_u = \int_0^s \frac{(s-u)^j}{j!}\, dY_u,$$

using $\phi(s) = 0$ and $Y_0 = 0$. □

Lemma 2 (Norm equivalence)**.** *For every $f \in L^2[0, T]$,*

$$\gamma^2 \|f\|^2 \;\leq\; \mathbb{E}\big[I(f)^2\big] \;\leq\; \big(\gamma^2 + h^2 T c_X\big) \|f\|^2.$$

Proof. Let $g(s) = f(T - s)$, so $\int_0^T g^2 = \|f\|^2$. By (2), $I(f) = hA + \gamma B$ with $A = \int_0^T g(s) X_s\, ds$ and $B = \int_0^T g(s)\, dV_s$. A is a functional of (X_0, W) and B of V, so they are independent. Since $\mathbb{E}B = 0$, $\mathbb{E}[AB] = 0$. The Itô isometry gives $\mathbb{E}B^2 = \|f\|^2$, hence $\mathbb{E}I(f)^2 = h^2 \mathbb{E}A^2 + \gamma^2 \|f\|^2$, which proves the lower bound. From (1), $X_t = e^{-at} X_0 + \sigma \int_0^t e^{-a(t-r)} dW_r$, so

$$\mathbb{E}X_t^2 = e^{-2at} P_\infty + \big(1 - e^{-2at}\big) \frac{\sigma^2}{2a} \leq c_X,$$

a convex combination of P_∞ and $\sigma^2/2a$. Minkowski's integral inequality and Cauchy–Schwarz then give $(\mathbb{E}A^2)^{1/2} \leq \int_0^T |g(s)| (\mathbb{E}X_s^2)^{1/2} ds \leq \sqrt{c_X} \int_0^T |g| \leq \sqrt{c_X T}\, \|f\|$, which proves the upper bound. □

Proposition 3 (Cohen et al., 2023, Thm. 1, constant-gain case)**.** $m_T := \mathbb{E}\big[X_T \mid \mathcal{F}^Y_T\big] = I(k) = \displaystyle\sum_{j \geq 0} K(-\lambda)^j\, S^{\mathtt{y}\mathtt{t}^j}_{0,T}$, *with the series converging in $L^2(\mathbb{P})$.*

Proof. The Kalman–Bucy filter (Kalman and Bucy, 1961; Bain and Crisan, 2009) for (1)–(2) is $dm_t = -a m_t\, dt + P_t h \gamma^{-2} (dY_t - h m_t\, dt)$, with $\dot{P}_t = \sigma^2 - 2aP_t - h^2 \gamma^{-2} P_t^2$, $m_0 = 0$ and $P_0 = P_\infty$. Since P_∞ is a fixed point of the Riccati equation, $P_t \equiv P_\infty$ and $dm_t = -\lambda m_t\, dt + K\, dY_t$. Applying Itô's product rule to $e^{\lambda t} m_t$ gives $m_T = \int_0^T K e^{-\lambda(T-s)} dY_s = I(k)$. Let $k_M(u) = K \sum_{j<M} (-\lambda u)^j / j!$. By Lemma 1 and linearity, $I(k_M) = \sum_{j<M} K(-\lambda)^j S^{\mathtt{y}\mathtt{t}^j}_{0,T}$. Taylor's theorem gives $\sup_{[0,T]} |k - k_M| \leq K(\lambda T)^M / M! \to 0$, so $\|k - k_M\| \to 0$, and by Lemma 2 $\mathbb{E}[(m_T - I(k_M))^2] \to 0$. □

4 Truncation with the exact coefficients

Let $m_T^{(M)} = \sum_{j<M} K(-\lambda)^j S_{0,T}^{\mathtt{yt}^j} = I(k_M)$, the representation truncated at signature level M, and $r_M = k - k_M$.

Theorem 4. *For every $M \geq 1$,*

$$\mathbb{E}\big[(m_T - m_T^{(M)})^2\big] \leq \big(\gamma^2 + h^2 T c_X\big)\, \frac{K^2 \lambda^{2M} T^{2M+1}}{(2M+1)(M!)^2}.$$

If moreover $M+1 > \lambda T$, then $\mathbb{E}[(m_T - m_T^{(M)})^2] \geq \gamma^2 \big(1 - \frac{\lambda T}{M+1}\big)^2 \frac{K^2 \lambda^{2M} T^{2M+1}}{(2M+1)(M!)^2}$.

Proof. For $x \geq 0$, the Lagrange form of the remainder gives $\big|e^{-x} - \sum_{j<M}(-x)^j/j!\big| = e^{-\xi} x^M/M! \leq x^M/M!$ for some $\xi \in (0,x)$. Hence $|r_M(u)| \leq K(\lambda u)^M/M!$, and $\|r_M\|^2 \leq K^2 \lambda^{2M} T^{2M+1}/\big((2M+1)(M!)^2\big)$. The upper bound follows from Lemma 2, since $m_T - m_T^{(M)} = I(r_M)$. For the lower bound, write $r_M(u)/K = \sum_{j \geq M} (-1)^j \tau_j$ with $\tau_j = x^j/j!$ and $x = \lambda u$. If $x < M+1$, then $\tau_{j+1}/\tau_j = x/(j+1) < 1$ for all $j \geq M$. The series is then alternating with decreasing terms, so $|r_M(u)|/K \geq \tau_M - \tau_{M+1} = \frac{x^M}{M!}\big(1 - \frac{x}{M+1}\big)$. For $u \leq T$ the last factor is at least $1 - \lambda T/(M+1) > 0$. Squaring, integrating and applying the lower half of Lemma 2 gives the claim. □

The bounds match up to constants, so the exact-coefficient error decays like $(\lambda T)^M/M!$. Their ratio between consecutive levels is $\rho_M^{\mathrm{T}} = \big(\lambda T/M\big)^2 \frac{2M-1}{2M+1}$.

5 Truncation with optimal coefficients

Let $\mathcal{S}_M = \operatorname{span}\{S_{0,T}^{\mathtt{yt}^j} : j < M\}$ and let $m_T^{(M,*)}$ be the $L^2(\mathbb{P})$-orthogonal projection of m_T onto $\mathcal{S}_M$. This is the best mean-square predictor built from these signature terms, and it is the quantity that least-squares regression on them estimates.

Theorem 5. *For every $M \geq 1$,*

$$\gamma^2 E_{M-1}(k)^2 \;\leq\; \mathbb{E}\big[(m_T - m_T^{(M,*)})^2\big] \;\leq\; \big(\gamma^2 + h^2 T c_X\big) E_{M-1}(k)^2.$$

Proof. By Lemma 1, $\mathcal{S}_M = \{I(p) : p \in \mathcal{P}_{M-1}\}$, since the $u^j/j!$ with $j < M$ span $\mathcal{P}_{M-1}$. As I is linear, $\mathbb{E}[(m_T - m_T^{(M,*)})^2] = \min_{p \in \mathcal{P}_{M-1}} \mathbb{E}[I(k-p)^2]$. Choosing p as the $L^2[0,T]$-best approximation of k and applying the upper half of Lemma 2 gives the upper bound. For any p, the lower half gives $\mathbb{E} I(k-p)^2 \geq \gamma^2 \|k-p\|^2 \geq \gamma^2 E_{M-1}(k)^2$. □

It remains to evaluate $E_n(k)$. Let i_ℓ denote the modified spherical Bessel function of the first kind, $i_\ell = \mathrm{i}_\ell^{(1)}$ in the notation of Olver et al. (2026, §10.47), and let $c = \lambda T/2$.

Lemma 6 (Best L^2 approximation of the kernel)**.** *For $n \geq 0$,*

$$E_n(k)^2 = K^2 e^{-2c}\, \frac{T}{2} \sum_{\ell > n} 2(2\ell+1)\, i_\ell(c)^2.$$

Moreover, with $\theta_\ell = 2(2\ell+1)\, c^{2\ell}/\big((2\ell+1)!!\big)^2$ and $q_n = c^2/\big((2n+3)(2n+5)\big)$, if $q_n < 1$ then

$$\theta_{n+1} \;\leq\; \frac{2}{T}\, K^{-2} e^{2c} E_n(k)^2 \;\leq\; \theta_{n+1}\, \frac{e^{c^2/(2n+5)}}{1 - q_n}.$$

Proof. The substitution $u = T(1-x)/2$ maps $x \in [-1,1]$ onto $u \in [0,T]$, maps $\mathcal{P}_n$ onto itself, and gives $k(u) = Ke^{-c}e^{cx}$ and $\|f\|^2 = \frac{T}{2}\int_{-1}^{1} f(u(x))^2 dx$. So $E_n(k) = Ke^{-c}\sqrt{T/2}\,\mathcal{E}_n$, where $\mathcal{E}_n$ is the best $L^2[-1,1]$ error of e^{cx} by $\mathcal{P}_n$.

Rayleigh's expansion (Olver et al., 2026, eq. 10.60.7) with $z = -\mathrm{i}c$, together with $\mathsf{i}_\ell^{(1)}(c) = \mathrm{i}^{-\ell}\mathsf{j}_\ell(\mathrm{i}c)$ (Olver et al., 2026, eq. 10.47.12) and the parity of j_ℓ, gives the Legendre expansion $e^{cx} = \sum_{\ell\geq 0}(2\ell+1)i_\ell(c)P_\ell(x)$. The Legendre polynomials are orthogonal with $\int_{-1}^{1} P_\ell^2 = 2/(2\ell+1)$, so the best approximation in $\mathcal{P}_n$ is the truncated series (Trefethen, 2019, Ch. 17), and $\mathcal{E}_n^2 = \sum_{\ell>n} 2(2\ell+1)i_\ell(c)^2$.

For the bounds, $i_\ell(c) = c^\ell \sum_{r\geq 0} \frac{(c^2/2)^r}{r!\,(2\ell+2r+1)!!}$ (Olver et al., 2026, eq. 10.53.3). All terms are positive, and $(2\ell+2r+1)!! \geq (2\ell+1)!!\,(2\ell+3)^r$, so

$$\frac{c^\ell}{(2\ell+1)!!} \leq i_\ell(c) \leq \frac{c^\ell}{(2\ell+1)!!}\, e^{c^2/(2(2\ell+3))}.$$

The lower bound follows by keeping only $\ell = n+1$. For the upper bound, $\theta_{\ell+1}/\theta_\ell = c^2/((2\ell+1)(2\ell+3)) \leq q_n$ for $\ell \geq n+1$, so $\sum_{\ell>n}\theta_\ell \leq \theta_{n+1}/(1-q_n)$. Also $e^{c^2/(2\ell+3)} \leq e^{c^2/(2n+5)}$ for $\ell \geq n+1$. □

Corollary 7. *With $n = M-1$, Theorem 5 and Lemma 6 give*

$$\gamma^2 K^2 e^{-\lambda T}\tfrac{T}{2}\,\theta_M \;\leq\; \mathbb{E}\big[(m_T - m_T^{(M,*)})^2\big] \;\leq\; \big(\gamma^2 + h^2 T c_X\big) K^2 e^{-\lambda T}\tfrac{T}{2}\,\theta_M\, \frac{e^{c^2/(2M+3)}}{1-q_{M-1}}.$$

Since $(2M+1)!! = (2M+1)!/(2^M M!)$ and $(2M)! \sim 4^M (M!)^2/\sqrt{\pi M}$, the root-mean-square error is of order $(\lambda T/4)^M/M!$ up to a factor polynomial in M. Compared with Theorem 4 this is smaller by a factor of order 4^{-M}. Consecutive levels shrink the mean-square error by $\rho_M^* = \theta_M/\theta_{M-1} = c^2/\big((2M-1)(2M+1)\big) \approx (\lambda T/4M)^2$.

6 Numerical verification

Setup.

- Parameters: $a = \sigma = h = 1$, $\gamma = 0.5$, $T = 1$. Hence $P_\infty = 0.3090$, $K = 1.2361$, $\lambda = \sqrt{5} = 2.2361$, $c_X = 0.5$, and $\gamma^2 + h^2 T c_X = 3\gamma^2$.
- Simulation: 6000 paths of (1)–(2) on an Euler grid with $dt = 10^{-3}$; 4000 for fitting, 2000 for testing.
- Signatures: computed with `esig` on the piecewise-linear interpolation of Y, and cross-checked against `pysiglib` (largest difference 2×10^{-16}). The target is $I(k)$ evaluated on that interpolated path, which the signature series represents exactly.
- Checks: the target agrees with a discrete-time Kalman filter to root-mean-square error 1.7×10^{-4}.

Errors and bounds are reported as mean-square error divided by $\mathrm{Var}(m_T) = 0.1651$ (test set). Code and raw output are in the repository (`d2_kb_signature.py`, `results/d2/summary.json`).

Exact coefficients (Table 1). Every measured error lies inside the bounds of Theorem 4. The level-to-level ratios approach ρ_M^{T} (0.070 against 0.069 at $M = 8$).

Table 1: Exact coefficients (Theorem 4). The lower bound needs $M + 1 > \lambda T$.

M	measured	lower bound	upper bound	measured ratio	ρ_M^{T}
1	1.87	–	1.16×10^{1}	–	–
2	1.65	1.88×10^{-1}	8.67	0.886	0.750
3	7.29×10^{-1}	2.23×10^{-1}	3.44	0.441	0.397
4	1.90×10^{-1}	8.52×10^{-2}	8.37×10^{-1}	0.261	0.243
5	3.26×10^{-2}	1.80×10^{-2}	1.37×10^{-1}	0.172	0.164
6	3.96×10^{-3}	2.48×10^{-3}	1.61×10^{-2}	0.122	0.118
7	3.59×10^{-4}	2.46×10^{-4}	1.42×10^{-3}	0.091	0.088
8	2.52×10^{-5}	1.85×10^{-5}	9.81×10^{-5}	0.070	0.069

Table 2: Optimal coefficients (Theorem 5, with $E_{M-1}(k)$ from Lemma 6).

M	measured	lower bound	upper bound	measured ratio	$\frac{E_{M-1}^2}{E_{M-2}^2}$	ρ_M^*
1	1.80×10^{-1}	1.42×10^{-1}	4.27×10^{-1}	–	–	–
2	1.17×10^{-2}	1.06×10^{-2}	3.18×10^{-2}	0.0645	0.0745	0.0833
3	3.72×10^{-4}	3.59×10^{-4}	1.08×10^{-3}	0.0320	0.0338	0.0357
4	7.02×10^{-6}	6.90×10^{-6}	2.07×10^{-5}	0.0189	0.0192	0.0198
5	8.38×10^{-8}	8.53×10^{-8}	2.56×10^{-7}	0.0119	0.0124	0.0126
6	7.31×10^{-10}	7.34×10^{-10}	2.20×10^{-9}	0.0087	0.0086	0.0087
7	4.68×10^{-12}	4.65×10^{-12}	1.40×10^{-11}	0.0064	0.0063	0.0064
8	2.10×10^{-14}	2.26×10^{-14}	6.79×10^{-14}	0.0045	0.0049	0.0049

Optimal coefficients (Table 2). Here the coefficients are fitted by least squares on the 4000 training paths. The measured errors sit almost exactly on the lower bound, so the observation-noise term dominates the constant in Lemma 2. Two entries ($M = 5, 8$) fall 1.7% and 7% below the lower bound. The bound concerns the continuous-time population error, while the table reports a 2000-path test-set estimate on a time grid; at $M = 8$ the relative error is also near 10^{-14}, where floating-point effects in the least-squares fit matter. The ratios match E_{M-1}^2/E_{M-2}^2 and approach $\rho_M^* \approx (\lambda T/4M)^2$. At $M = 8$ the error is about 10^9 times smaller than with the exact coefficients.

7 Outlook: a nonlinear filter

For the Beneš model $dX_t = \tanh(X_t)\,dt + dW_t$, $dY_t = X_t\,dt + \gamma\,dV_t$, $X_0 = 0$ (Beneš, 1981), the filter is finite-dimensional. Since $\tanh' + \tanh^2 = 1$, Girsanov's theorem gives the posterior as $\cosh(x)$ times the Gaussian Kalman–Bucy posterior of a Brownian signal. Its mean is therefore $m_T + P_T \tanh(m_T)$, where $m_T = \int_0^T \sinh(hs/\gamma)\,dY_s/(\gamma\cosh(hT/\gamma))$ and $P_T = (\gamma/h)\tanh(hT/\gamma)$. Again m_T is a single-$\mathtt{y}$ signature functional.

In simulations with $\gamma = 1.5$ and $T = 2$ (6000 paths; the closed form agrees with a 20,000-particle filter to root-mean-square error 0.014 against a target standard deviation of 1.91):

- Least squares on the words $\mathtt{yt}^j$ stops at 1.25×10^{-2} (relative mean-square error) from $M = 3$ onwards. This is exactly the share of the target's variance that is nonlinear in m_T.
- Ridge regression on the full truncated signature goes below this floor: 6.6×10^{-3}, 2.6×10^{-3},

1.7×10^{-3} and 7.8×10^{-4} for $M = 3, \ldots, 6$. That is roughly a factor of 2 per level. For $M \geq 7$ the error is limited by the training-set size.

Bounding this slower, geometric rate requires words with several $\mathtt{y}$, where the Itô and Stratonovich lifts differ, and polynomial approximation of tanh composed with a Gaussian functional. This is left for future work.

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